Firoozbakht's Conjecture: A Structured Attack

A machine-checked reduction, an unconditional finite-range architecture, the closure of the Riemann-Hypothesis route, and what remains open

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Abstract

Firoozbakht's conjecture asserts that $p_{n+1}^{1/(n+1)} < p_n^{1/n}$ for every $n \geq 1$, i.e. that $n \mapsto p_n^{1/n}$ is strictly decreasing. It is open. This paper reports a structured attack on it that neither proved nor refuted it, and states precisely what the attack did establish. Five things. (i) The reduction of the conjecture to the prime-gap inequality $g_n < T_n$, where $T_n := p_n(p_n^{1/n} - 1)$, is *machine-checked* in Lean 4 against Mathlib: the development builds with exactly one `sorryAx` dependent, namely the conjecture itself. (ii) The maximal-gap pruning rule on which the only computationally live verification route depends is licensed by an elementary *monotone-bar principle*, which shows that the obligation the attack had ranked first — monotonicity of the exact bar T — is misdirected: the pruning may be run against Kourbatov's monotone surrogate $S(x) = \log^2 x - \log x - 1.17$, whose oscillation is nil. (iii) An unconditional finite-range theorem, resting on a single explicit estimate for $\pi(x)$ due to Dusart, converts any first-occurrence prime-gap table into a proof of the conjecture up to that table's reach; it reproduces the published constant 1920 at 2^{64} with no tuning, and it is accompanied by a table-free window $396\,738 \leq p_n \leq 777\,600$ together with a proof that no presently known unconditional gap bound can extend that window. (iv) Four theorems close the Riemann-Hypothesis route *as a route*: the sharpest published RH-conditional gap bound certifies the Firoozbakht inequality at exactly one index in its range of availability, and no envelope of power type at any positive exponent can certify it beyond finitely many indices. (v) An independent exhaustive sweep to 10^{11} ($4\,118\,054\,812$ consecutive prime pairs) finds no violation and no near-miss, with $\max_{n \geq 10} g_n/T_n = 0.8318$. We also state, at length, what the attack did *not* establish, including two named unrepaired defects in the underlying corpus and the fact that the citation audit for this paper has not been run.

Contents

1 Introduction

1.1 The conjecture

Let $p_1 = 2 < p_2 = 3 < p_3 = 5 < \dots$ enumerate the primes, one-indexed.

Definition 1.1 (Firoozbakht's conjecture, F)**.** F is the assertion

$$p_{n+1}^{1/(n+1)} < p_n^{1/n} \qquad \text{for every } n \geq 1, \tag{1}$$

equivalently: the sequence $\left(p_n^{1/n}\right)_{n\geq 1}$ is strictly decreasing.

The conjecture is due to Farideh Firoozbakht (University of Isfahan, 1982) and was never published by her [Fir82]; its first appearance in print is [Rib04, p. 185]. Its date and unpublished status are attested independently by [Kou15a, §1], [FM18, §1] and [Vis19]. Sun states it in the form "the sequence $(\sqrt[n]{p_n})_{n\geq 1}$ is strictly decreasing" [Sun13, §1]; Kourbatov states the equivalent $p_{k+1} < p_k^{1+1/k}$ [Kou15a, §1, eq. (1)]. All of these are the same statement, and we use them interchangeably.

1.2 Why it is interesting, and why it is hard

F is a statement about prime gaps in disguise. Writing $g_n := p_{n+1} - p_n$ and $L_n := \log p_n$ (natural logarithm throughout), F is exactly the assertion that g_n stays below the threshold $T_n := p_n(p_n^{1/n} - 1)$ at every index (Lemma 2.1), and $T_n = L_n^2 - L_n - 1 + o(1)$ [Kou15a, §5, Thm. 5]. So F implies a Cramér-scale gap bound with the leading constant 1:

$$\mathsf{F} \implies \big(\, g_k < L_k^2 - L_k - 1 \quad \text{for all } k > 9\,\big), \tag{2}$$

which is [Kou15a, §2, Thm. 1]; see also [FM18, Thm. 2.2] and the weaker +1 variant in [Sun13, §1]. In the converse direction,

$$\big(\, g_k < L_k^2 - L_k - 1.17 \quad \text{for all } k > 9,\ p_k \geq 29\,\big) \implies \mathsf{F}, \tag{3}$$

which is [Kou15a, §4, Thm. 3]. Thus F is trapped between two $\log^2$-scale gap bounds differing by an additive 0.17.

That sandwich is the source of the difficulty. The best *unconditional* prime-gap bound available is $g_n = O\big(p_n^{0.525}\big)$ [BHP01]; under the Riemann Hypothesis the state of the art is $g_n \leq \frac{22}{25}\sqrt{p_n}\,\log p_n$ for $p_n > 3$ [CMS19, §1.2, Thm. 5]. Both are *powers* of p_n; F needs a *polylogarithmic* bound. The distance is square-root scale against log scale, and no known method bridges it, even conditionally on RH. Section 5 turns this observation into theorems.

On the other side, the standard heuristic points the other way. Cramér's random model, in the corrected form Granville gives it, suggests

$$\max_{p_n \leq x}(p_{n+1} - p_n) \ \gtrapprox\ 2e^{-\gamma}\log^2 x, \qquad 2e^{-\gamma} = 1.12292\ldots,$$

[Gra95, preprint p. 12, after eq. (20)], which is *incompatible* with F via (2). At least one of the two — Firoozbakht's conjecture or the Cramér–Granville heuristic — must fail, and no present technique says which. This is the single most important framing sentence of the subject, and we return to it in §9.

1.3 Literature

The relevant literature falls into five groups.

Statement, strengthenings and consequences. Ribenboim records the conjecture [Rib04, p. 185]; Sun uses it as the motivating contrast for a provable analogue about sums of primes [Sun13, Thm. 2.1]. Ferreira and Mariano derive a systematic list of consequences [FM18]: the necessary gap bound ([FM18, Thm. 2.2]), $g_n < \sqrt{n}$ for $n \geq 3645$ ([FM18, Lem. 3.2]), a Sierpiński-type consequence about primes in rows of the $n \times n$ array of $1, \ldots, n^2$ ([FM18, Cons. 3.3]), and — most importantly for us — the only unconditional positive statement in the literature about F itself, Theorem 2.4 below. Visser [Vis19] states and verifies the two known strengthenings, Nicholson's (traced to an OEIS comment [Slob], otherwise unpublished) and Farhadian's [FJ17], and records the implication chain Farhadian ⇒ Nicholson ⇒ Firoozbakht [Vis19, Conj. 2], proved in [FM18, Thm. 4.5] together with Forgues' weaker variant.

Criteria. The necessary and sufficient conditions (2) and (3) are Kourbatov's [Kou15a, §2 Thm. 1, §4 Thm. 3], together with a family of sufficient conditions with $b \to 1$ [Kou15a, §4, Thm. 4] that presuppose the finite verification (they are stated for $p_k > 4 \cdot 10^{18}$). Both rest on Axler's effective bounds for $\pi(x)$ [Axl14]; see §10 for why that dependency is load-bearing and unresolved here.

Verification. Kourbatov verified F for all $p_k < 4 \cdot 10^{18}$ [Kou15b, §4, Thm.], using the first-occurrence prime-gap table of Oliveira e Silva, Herzog and Pardi [OHP14]; the 2023 endnotes to [Kou15b] extend the range to $2^{64} \approx 1.8447 \cdot 10^{19}$ and record that "prime gaps of size $g < 1920$ cannot violate" the conjecture. Visser independently reports verification up to the 81st maximal prime gap, "certainly for all primes $p < 2^{64}$" [Vis19, Abstract, §1]. We reconstruct the architecture of this verification from first principles in §4.

The heuristic side. Cramér proves $\limsup(P_{n+1} - P_n)/(\log P_n)^2 = 1$ with probability 1 for his urn model and only *suggests* the analogue for the primes [Cra36, pp. 24, 26–27]; this distinction is not a pedantry and we preserve it everywhere. Granville corrects the model and obtains the opposite prediction [Gra95, preprint pp. 10, 12]. The empirical record of the Cramér–Shanks–Granville ratio $(q-p)/\log^2 p$ is A111943 [Sloa]; Shanks conjectured the ratio never reaches 1 [Sha64], Granville the opposite. The record stands at 0.9206 at $p = 1\,693\,182\,318\,746\,371$ (gap 1132, B. Nyman, 1999) [Sloa].

Large gaps. The best lower bound on the largest gap below X is

$$G(X) \;\geq\; f(X)\,\frac{\log X \;\log\log X \;\log\log\log\log X}{(\log\log\log X)^2}, \qquad f(X) \to \infty,$$

[For+16] — an iterated-log factor above $\log X$, but still a full power of log below the $\log^2 X$ a counterexample to F would need.

1.4 What this paper claims, and the vocabulary of its claims

This paper is an honest report on an attack that did not settle the conjecture. To keep the strength of each statement legible we fix the following vocabulary and use it without exception.

Code	Meaning
[K]	**Machine-checked** by the Lean 4 kernel against Mathlib (§8).
[P]	Paper proof, derived in this work, independently re-derived by an adversarial review leg.
[P·s]	As [P], but resting on a cited source not opened in this work.
[C]	Finite computation, exhaustive, independently reproduced by two code paths.
[H]	Heuristic. Not evidence for or against anything, and never used in a proof.

The word *proved*, unqualified, is reserved in this paper for statements carrying [K]. Everything at [P] or [P·s] is a paper proof and is labelled as such at the point of use. In particular:

F is neither proved nor refuted in this paper, and nothing below should be read as evidence that either is close. The defensible summary sentence is: *Firoozbakht's conjecture is numerically robust over the verified range and simultaneously incompatible with the standard Cramér–Granville heuristic; at least one of the two must fail, and no current technique can say which.*

1.5 Contributions

1. [K] A Lean 4/Mathlib formalisation in which the reduction $\mathsf{F} \leftrightarrow (\forall n \geq 1,\ g_n < T_n)$ is machine-checked, with an exhaustive list-free axiom audit over 60 declarations finding exactly one `sorryAx` dependent — the conjecture itself (§8).
2. [P] The monotone-bar principle (Lemma 3.1) and its truncated form, which discharge the practical content of the attack's ranked-first open obligation without proving that obligation, and show it to be misdirected (§3).
3. [P·s], [C] An unconditional finite-range theorem (Theorem 4.4) resting on a single explicit $\pi(x)$ estimate, reproducing the published constant 1920 at 2^{64}; plus a table-free window and a proof that it closes permanently (§4).
4. [P] Four theorems closing the RH route as a route (§5), including a Lambert-W determination of the critical constant $2/e$ and an explicit counter-model showing that Cramér's $\limsup$ hypothesis does not entail F over integer sequences.
5. [C] An independent exhaustive verification to 10^{11} with a verification discipline that raises rather than returns when a margin lands inside its error budget (§7).
6. An explicit account of what is *not* established, including two unrepaired defects in the underlying corpus and the absent citation audit (§10).

2 Setup

2.1 Notation

Throughout, p_n is the n-th prime with $p_1 = 2$ (one-indexed), $\pi(x)$ is the prime-counting function, so that

$$\pi(p_n) = n, \tag{4}$$

and

$$L_n := \log p_n, \qquad g_n := p_{n+1} - p_n, \qquad T_n := p_n\big(p_n^{1/n} - 1\big) = p_n\big(e^{L_n/n} - 1\big).$$

All logarithms are natural. We write $\rho_n := g_n/T_n$ for the *normalised objective*; F at n is exactly $\rho_n < 1$.

Identity (4) looks like bookkeeping and is not: it is the hinge on which every analytic estimate in this paper turns, because T_n depends on the pair (p_n, n) and the only way to control n analytically is through π.

2.2 The reduction

Lemma 2.1 (Gap form of F; [K])**.** *For every $n \geq 1$,*

$$p_{n+1}^{1/(n+1)} < p_n^{1/n} \iff p_{n+1}^{n} < p_n^{n+1} \iff g_n < T_n.$$

Proof. Idea: Raise to the power $n(n+1)$, take logarithms, and subtract p_n from both sides; every step is an equivalence because the maps involved are strictly increasing.

All quantities are real and exceed 1. The map $t \mapsto t^{n(n+1)}$ is strictly increasing on $(0, \infty)$, so raising $p_{n+1}^{1/(n+1)} < p_n^{1/n}$ to that power gives $p_{n+1}^n < p_n^{n+1}$, and the step is reversible. Applying the strictly increasing log gives $n \log p_{n+1} < (n+1) \log p_n$, i.e. $\log p_{n+1} < L_n(1 + 1/n)$, i.e. $p_{n+1} < p_n e^{L_n/n}$, i.e. $p_n + g_n < p_n + p_n(e^{L_n/n} - 1)$. Subtracting p_n gives $g_n < T_n$. □

Lemma 2.1 is the one result in this paper carrying [K]: it is formalised and machine-checked in Lean 4 (§8), together with the intermediate equivalence, and both equivalences are pinned to numerals ($g_1 = 1$, $T_1 = 2^2 - 2$) so that a mis-stated g or T fails to compile rather than passing quietly.

Remark 2.2 (Strict versus non-strict). Visser states the conjecture with $\leq$ [Vis19, Conj. 3, eq. (2.4)]. The two forms coincide if and only if T_n is never an integer. That is true for $n \geq 2$: $p_n^{1/n}$ rational would force it to be an integer and p_n to be a perfect n-th power, which unique factorisation forbids for a prime. We nonetheless use the strict form throughout and never silently convert between them.

2.3 The two known unconditional facts about F

Theorem 2.3 (Verified range; [C], cited)**.** *F holds for every n with $p_n < 2^{64} = 18\,446\,744\,073\,709\,551\,616 \approx 1.8447 \cdot 10^{19}$.*

This is [Kou15b, endnotes, 5 Jan 2023], corroborated by [Vis19, Abstract, §1 eq. (1.4)]. Its computational input is the first-occurrence gap table of [OHP14]. *This paper does not re-establish Theorem 2.3*; §4 reconstructs the implication from a table to the range, and §10 records that the table itself was never opened in this work.

Theorem 2.4 (Infinitely often; cited)**.** *There are infinitely many $n \in \mathbb{N}$ with $p_n^{1/n} > p_{n+1}^{1/(n+1)}$.*

This is [FM18, Thm. 5.2], proved unconditionally there via Zhang's bounded-gaps theorem (cited as [FM18, Thm. 5.1]). It is the only unconditional theorem in the literature that says something positive about F itself.

Caveat 2.5. Theorem 2.4 says *infinitely often*, not *eventually*. It is fully compatible with infinitely many failures. It must never drift into "F holds for large n".

3 The monotone-bar principle and the maximal-gap reduction

3.1 The obligation, and why it is the wrong one

Every practical verification of F over a range checks *record* gaps rather than all indices. The standard justification for that pruning is the following statement, which the present attack initially ranked as its most tractable open obligation:

> **(P6′)** For $m < n$ straddling a record gap, $T_m \leq T_n$.

(P6′) is a statement about the oscillation of the exact bar T, and T is *not* monotone: over the primes below $3 \cdot 10^6$ it decreases at about 55.9% of steps (§7). The main finding of this section is that no consumer of (P6′) needs it.

Lemma 3.1 (Monotone-bar principle; [P])**.** *Let B be a real-valued function of the prime p, nondecreasing in p. Call k a B-*breach *if $g_k \geq B(p_k)$. If a B-breach exists, then the least one, k_0, satisfies $g_m < g_{k_0}$ for every $m < k_0$; that is, k_0 is a strict maximal-gap index.*

Proof. Idea: Minimality gives an upper bound at every earlier index; monotonicity moves that bound forward to k_0; the two chain.

Let $m < k_0$. Then $p_m < p_{k_0}$, so $B(p_m) \leq B(p_{k_0})$ by monotonicity. By minimality of k_0, the index m is not a B-breach, so $g_m < B(p_m)$. Chaining, $g_m < B(p_m) \leq B(p_{k_0}) \leq g_{k_0}$. □

Only monotonicity of B is used: no arithmetic, no analytic input, no property of the primes beyond $p_m < p_{k_0}$. The conclusion is strict, which is exactly the difference between "the first failure carries a record gap" and "the first failure carries a gap at least as large as all earlier ones". The lemma says nothing about the *second* breach, and it should not: breaches after the first need not be records.

Explicit bars have validity ranges, and the first few primes sit outside them, so we need a truncated form.

Lemma 3.2 (Truncated monotone-bar principle; [P]). *Let B be nondecreasing and let $1 \le N_1 \le N_2$. Suppose*

(i) $g_m < B(p_m)$ *for every maximal-gap index m with $N_1 \le m \le N_2$;*

(ii) $\max\{g_m : m < N_1\} < B(p_{N_1})$.

Then $g_k < B(p_k)$ for every k with $N_1 \le k \le N_2$.

Proof. Idea: Run Lemma 3.1 inside the window, and let hypothesis (ii) *absorb everything below it in one step.*

Suppose not, and let k_0 be least in $[N_1, N_2]$ with $g_{k_0} \ge B(p_{k_0})$. Take any $m < k_0$. If $m \ge N_1$, minimality gives $g_m < B(p_m) \le B(p_{k_0}) \le g_{k_0}$. If $m < N_1$, then by (ii) and monotonicity $g_m \le \max\{g_j : j < N_1\} < B(p_{N_1}) \le B(p_{k_0}) \le g_{k_0}$. Either way $g_m < g_{k_0}$, so k_0 is a maximal-gap index in $[N_1, N_2]$ breaching B, contradicting (i). □

3.2 Instantiating at Kourbatov's monotone surrogate

Put

$$S(x) := \log^2 x - \log x - 1.17.$$

Lemma 3.3 ([P]). *S is strictly increasing on $x > e^{1/2}$, hence at every prime.*

Proof. Idea: One derivative.

$dS/dx = (2\log x - 1)/x > 0$ for $\log x > 1/2$, and $e^{1/2} = 1.6487\ldots < 2$. □

Lemma 3.4 (S lies below the true bar; [P·s]). *For every n with $p_n \ge 2\,634\,800\,823$ we have $T_n > S(p_n)$.*

Proof. Idea: Replace the index n by an upper bound on $\pi(p_n)$, then use $e^u - 1 > u$.

By (4), $n = \pi(p_n)$. Axler's Corollary 3.5, in the corrigendum-corrected form, states $\log x - 1 - 1.17/\log x < x/\pi(x)$ for $x \ge 2\,634\,800\,823$ [Axl14]. With $x = p_n$ and $\ell = L_n$ this gives $\pi(x) < x/(\ell - 1 - 1.17/\ell)$, hence $u := \ell/\pi(x) > (\ell^2 - \ell - 1.17)/x$, and therefore $T_n = x(e^u - 1) > xu > \ell^2 - \ell - 1.17 = S(p_n)$. □

Caveat 3.5 (The dependency, named). Lemma 3.4 rests on [Axl14], which *was not opened in this work*; it is quoted through Kourbatov's proofs. Axler's own corrigendum moved the validity range of Corollary 3.5 from $x \ge 5.43$ to $x \ge 2\,634\,800\,823$ — nine orders of magnitude — so the range is load-bearing, not decorative: in-run data below $3 \cdot 10^6$ exhibits 14 426 indices with $T_n \le S(p_n)$. Consequently Lemma 3.4 and Theorem 3.6 carry [P·s], not [P]. See §10 and Remark 3.9 for the Dusart-only fallback.

Theorem 3.6 (Maximal-gap reduction; [P·s]). *Let $N_2 \ge 10$ and suppose $g_m < S(p_m)$ for every maximal-gap index $m \in [10, N_2]$. Then $g_k < S(p_k)$ for every $k \in [10, N_2]$, and consequently* F *holds at every $k \in [10, N_2]$ with $p_k \ge 2\,634\,800\,823$.*

Proof. Idea: Lemma 3.2 with the monotone surrogate S in place of the oscillating T; the small-index hypothesis is one finite comparison.

Apply Lemma 3.2 with $B = S$ (nondecreasing by Lemma 3.3), $N_1 = 10$ and N_2 as given. Hypothesis (i) is the assumption. Hypothesis (ii) is the finite fact $\max\{g_j : j \le 9\} = 6 < 6.80139\ldots = S(29) = S(p_{10})$, verified in exact arithmetic (§7, item V2). Hence $g_k < S(p_k)$ throughout $[10, N_2]$. Where $p_k \ge 2\,634\,800\,823$, Lemma 3.4 gives $S(p_k) < T_k$, so $g_k < T_k$, which is F at k by Lemma 2.1. □

Remark 3.7 (The obligation was misdirected). Theorem 3.6 discharges the entire practical content of (P6′) without proving (P6′). The oscillation of T is irrelevant to the pruning, because the pruning never runs against T: it runs against S, which does not oscillate. The pieces were all present in the attack's own concept-card set — the bar S, the (unstated but immediate) monotonicity of S, and the open obligation — and were never joined. The join is Lemma 3.2.

Remark 3.8 (Range discipline). Theorem 3.6 delivers nothing below $p_k = 2\,634\,800\,823$, where Lemma 3.4 is unavailable and in fact false. That segment is covered by direct computation, i.e. by Theorem 2.3, and the two must be joined explicitly, never assumed to overlap.

Remark 3.9 (The Dusart-only fallback). The Axler dependency of Lemma 3.4 can be removed entirely at the cost of a looser bar: Lemma 4.2 below proves $T_n > B(p_n)$ with $B(x) = \log^2 x - 1.1 \log x$ for $p_n \ge 60\,184$, from Dusart alone, and B is monotone by Lemma 4.3. Substituting B for S throughout Theorem 3.6 gives a fully [P] version, with a bar looser by roughly $0.1L$ — a relative deficit measured at 0.193% at $p = 2{\cdot}10^7$ and 0.196% at $p = 1.69{\cdot}10^{15}$. We state the Axler version because it is the one the literature uses; we flag the fallback because it is the one that needs no unopened source.

3.3 Where the exact bar still resists

Lemma 3.1 does not settle the statement "if F fails, its first failure carries a record gap" against the *exact* bar T, because T is not monotone. What can be proved unconditionally is a windowed version.

Theorem 3.10 (Near-record maximality, Dusart only; [P])**.** *Suppose F is false and let n_0 be its least failure. By Theorem 2.3, $p_{n_0} > 2^{64}$, hence $L_{n_0} > 44.36$ and $T_{n_0} > 1919$. Let $m < n_0$ with*

$$p_m \;\le\; p_{n_0}\, e^{-0.0623} \;=\; 0.93960\cdots p_{n_0}.$$

Then $g_m < g_{n_0}$.

Proof. Idea: Bound T_m above and T_{n_0} below with Dusart's two-sided $\pi(x)$ estimates and ask how far apart the two logarithms must be for the upper bound at m to fall below the lower bound at n_0; the answer is a constant.

If $p_m < 60\,184$ then $g_m \le 72$ (the largest gap below 60 184; §7, item V1) while $g_{n_0} \ge T_{n_0} > 1919$, and we are done. Otherwise $p_m \ge 60\,184 > 5393$, so both halves of [Dus10, Thm. 6.9, eq. (6.6)] apply. Put $\ell = L_m \ge \log 60\,184 = 11.005$, $\lambda = L_{n_0} = \ell + d$ and $\varepsilon := (\ell^2 - \ell)^2/p_m$. It suffices that $(\ell^2 - \ell)\big(1 + (\ell^2 - \ell)/p_m\big) \le \lambda^2 - 1.1\lambda$, i.e. that

$$2\ell d + d^2 - 1.1d - 0.1\ell \;\ge\; \varepsilon.$$

The left side is increasing in $d \ge 0$, so it suffices that $d \ge d^*(\ell) := (0.1\ell + \varepsilon)/(2\ell - 1.1)$. Since $p_m \ge \max(e^\ell, 60\,184) = e^\ell$ for $\ell \ge \ell_0 := \log 60\,184 = 11.00516$, we have $\varepsilon(\ell) \le (\ell^2 - \ell)^2 e^{-\ell}$ there; that majorant has logarithmic derivative $2(2\ell-1)/(\ell^2-\ell)-1$, equal to $-0.618 < 0$ at ℓ_0 and decreasing thereafter, so $\varepsilon(\ell) \le \varepsilon(\ell_0) = 0.20145$. Hence $d^*(\ell) \le (0.1\ell + 0.20145)/(2\ell - 1.1)$, whose numerator derivative test $0.1(2\ell - 1.1) - 2(0.1\ell + 0.20145) = -0.5129 < 0$ shows it is decreasing, so it is maximised at ℓ_0, where it equals 0.062264. Thus $d \geq 0.0623$ suffices for every admissible ℓ. $\square$

Remark 3.11 (The Axler-sharpened version is withheld). A companion statement sharpens the exponent 0.0623 by roughly a factor of ten under Axler's corollaries. **We deliberately do not state its constant.** An adversarial review of the underlying corpus found that the sharpened version's derivation misapplies its upper bound on T_m by a factor of order ℓ^2, and that the in-run numerical check reproduced the error rather than catching it — a check written from a derivation cannot catch an error in the derivation. The review also established that the sharpened statement's *conclusion* survives under the corrected form of the bound, but the repair has not been applied to the source document. This paper therefore quotes only Theorem 3.10, whose constant was independently confirmed conservative and which uses no unopened source. See §10.

3.4 The obstruction inside the residual window

Theorem 3.10 leaves a sliver: the primes p_m within a factor $e^{-0.0623}$ of p_{n_0}. The obstruction there is exact and worth naming, because it explains why no sharpening of constants closes the window.

To first order in y/p_m, with $y = p_n - p_m$ and $k = \pi(p_n) - \pi(p_m)$,

$$T_m \leq T_n \quad \iff \quad k \ \leq \ y\Big(1 + \frac{1}{L_m}\Big)\frac{\pi(p_m)}{p_m}\big(1 + O(y/p_m)\big),$$

i.e., since $\pi(x)/x \approx 1/(\log x - 1)$: *$T_m \leq T_n$ holds exactly when the interval $(p_m, p_n]$ contains no more than about $y/(L-2)$ primes.* The surplus $1/(L-1) - 1/L \approx 1/L^2$ between the running-average density and the local density is the entire drift that makes T coarsely increase.

Proposition 3.12 (Obstruction; [P]). *Closing the residual window of Theorem 3.10 requires an unconditional upper bound on the number of primes in an interval of length y of order $x/\log x$ that is within a factor $1 + 2/L$ of the truth. The unconditionally available bound of Brun–Titchmarsh type gives only a factor of about* 2.

The required relative slack over the prime-number-theorem count is 4.7% at 2^{64} and shrinks to 1% at $L = 200$, while the Brun–Titchmarsh-to-needed ratio stays near 2.1 across that range. So the needed estimate is not a constant improvement but an asymptotically exact short-interval count. *This is, in our judgement, the most tractable genuinely open analytic node the attack isolated — and it is still an open problem in analytic number theory.*

Caveat 3.13 (Unsourced, and used only negatively). Neither the Brun–Titchmarsh inequality nor the second Hardy–Littlewood conjecture is covered by a source-ledger row in this work. Both are standard and undisputed; they are used here only to *name* an obstruction, never to support a positive claim. Proposition 3.12 is stated at [P] for its criterion ($1 + 2/L$, derived above) and its comparison to Brun–Titchmarsh is flagged as unsourced.

4 The unconditional finite-range architecture

The published frontier 2^{64} is not a per-pair sweep: it is a theorem about a table. This section reconstructs that theorem from first principles, with explicit constants, resting on *one* cited inequality — and in particular avoiding the Axler chain of Caveat 3.5.

4.1 The analytic input

Lemma 4.1 (Dusart's upper bound for π; cited, tier L0). *$\pi(x) \le x/(\log x - 1.1)$ for all $x \ge 60\,184$.*

This is [Dus10, Thm. 6.9, eq. (6.6)], read at the locator. It is the only external mathematical input to Theorem 4.4.

Lemma 4.2 (Explicit floor under the bar; [P]). *Set $B(x) := \log^2 x - 1.1 \log x$. Then $T_n > B(p_n)$ for every n with $p_n \ge 60\,184$.*

Proof. Idea: The only opaque ingredient of T_n is the index $n = \pi(p_n)$; an upper *bound on π turns the bar into a lower bound in $\log p_n$ alone.*

Since $e^t - 1 > t$ for $t > 0$ and $L_n/n > 0$, we have $T_n > p_n L_n/n$. By (4), $n = \pi(p_n)$, and Lemma 4.1 at $x = p_n \ge 60\,184$ gives $n \le p_n/(L_n - 1.1)$, with $L_n - 1.1 > 0$ in this range. Hence $p_n/n \ge L_n - 1.1$ and

$$T_n \;>\; L_n \cdot \frac{p_n}{n} \;\ge\; L_n(L_n - 1.1) \;=\; B(p_n). \qquad \square$$

The direction is the content. T_n *decreases* in n at fixed p_n, so lower-bounding T_n requires an upper bound on the rank, which is precisely what Dusart's upper bound on π supplies. The validity range is load-bearing and not decoration: $L(L - 1.1)$ crosses $L^2 - L - 1$ — the true asymptotic size of T_n — exactly at $L = 10$, i.e. $x = e^{10} = 22\,026$, below which Lemma 4.2 is false. Dusart's threshold 60 184 clears that crossing, but only by about 0.10 in L.

Lemma 4.3 ([P]). *B is strictly increasing on $[e^{0.55}, \infty) \supseteq [2, \infty)$.*

Proof. Idea: One derivative, in the variable L.

With $L = \log x$, $dB/dL = 2L - 1.1 > 0$ for $L > 0.55$, and $L \mapsto \log x$ is strictly increasing; $e^{0.55} = 1.7333 < 2$. $\square$

Lemma 4.3 is the whole point: B is a function of p *alone*, not of the pair (p_n, n), so the non-monotone object T never appears in a comparison between two different indices. In particular this section is independent of (P6′), and any downstream argument that gates the verified range on (P6′) is importing a dependency that is not there.

4.2 The finite-range theorem

Theorem 4.4 (Unconditional finite-range verification; [P]). *Let $X_0 := 60\,184$ and let $X \ge X_0$. Suppose:*

(H1) *$p_{n+1}^{n} < p_n^{n+1}$ for every n with $p_n < X_0$;*

(H2) *a table is available listing, for every gap value g occurring as $g_n = g$ for some n with $p_n \le X$, the first occurrence $q(g) := \min\{p_n : g_n = g\}$;*

(H3) *for every gap value g in that table with $q(g) \ge X_0$, one has $g \le B(q(g))$.*

Then F holds at every n with $p_n \le X$.

Proof. Idea: A gap that was already safe where it first *occurred is safe everywhere later, because the floor B only rises; gaps whose first occurrence is below X_0 are all too small to matter.*

Let n satisfy $p_n \le X$. If $p_n < X_0$, then (H1) and Lemma 2.1 give F at n. Otherwise $p_n \ge X_0$. Put $g := g_n$ and $q := q(g) \le p_n$.

If $q \geq X_0$: by (H3), $g \leq B(q)$; by Lemma 4.3 and $q \leq p_n$, $B(q) \leq B(p_n)$; by Lemma 4.2, $B(p_n) < T_n$. Hence $g_n < T_n$, and Lemma 2.1 gives F at n.

If $q < X_0 \leq p_n$: the first occurrence of g lies in the base-case range, so $g \leq G_0 := \max\{g_k : p_k < X_0\} = 72$, attained at $p = 31\,397$ (§7, item V1, exact integer arithmetic over the 6 076 gaps below X_0). Since $B(X_0) = B(60\,184) = 109.008\ldots > 72 \geq g$, Lemmas 4.3 and 4.2 give $g \leq G_0 < B(X_0) \leq B(p_n) < T_n$, so F at n. $\square$

Corollary 4.5 (Decision procedure; [P])**.** *For a gap value g define*

$$S^\sharp(g) \;:=\; \exp\!\left(\frac{1.1 + \sqrt{1.21 + 4g}}{2}\right),$$

the unique p with $B(p) = g$, so that $g \leq B(p) \iff p \geq S^\sharp(g)$. Then hypothesis (H3) *reads $q(g) \geq S^\sharp(g)$, and Theorem 4.4 says:* a gap of size g occurring at a prime $p \geq \max(S^\sharp(g), 60\,184)$ cannot violate F.

Proof. Idea: Invert the quadratic $B(p) = \log^2 p - 1.1 \log p$ in the variable $\log p$ and use Lemma 4.3.

Solving $L^2 - 1.1L = g$ for $L > 0$ gives $L = (1.1 + \sqrt{1.21 + 4g})/2$; B is strictly increasing (Lemma 4.3), so $g \leq B(p)$ iff $\log p$ exceeds that root. The restatement of (H3) is immediate, and the displayed sentence is the second case analysis of Theorem 4.4 read backwards. $\square$

Note that a gap size which passes is safe *everywhere*, not merely below X: $S^\sharp(g)$ does not depend on X. The verification of a range therefore costs one finite integer check below 60 184, plus one comparison per distinct gap value. Nothing else.

4.3 Independent reproduction of the published constant 1920

Proposition 4.6 ([C])**.** $B(2^{64}) = L(L - 1.1)\big|_{L=\log 2^{64}} = 1919.1379834975\ldots$, *so a gap of at least* 1920 *is required to violate* F *at a prime just below* 2^{64}*. Equivalently, the largest even g with $S^\sharp(g) \leq 2^{64}$ is $g = 1918$: $S^\sharp(1918) = 1.8209 \cdot 10^{19} \leq 2^{64} = 1.8447 \cdot 10^{19} < S^\sharp(1920) = 1.8629 \cdot 10^{19}$.*

Kourbatov's 2023 endnote states that "prime gaps of size $g < 1920$ cannot violate" the conjecture [Kou15b]; since every prime gap after $g_1 = 1$ is even, $g < 1920$ means exactly $g \leq 1918$. The published integer falls out of Lemma 4.2 with no tuning.

Caveat 4.7 (What the agreement is worth, and what it is not)**.** Three caveats, all stated rather than glossed. (a) The two derivations are not independent in their analytic ingredient: both replace n by an upper bound on $\pi(p_n)$; they differ in *which* bound (Dusart eq. (6.6) here, the Axler-based $L^2 - L - 1.17$ there). What the agreement does establish is that no sign error, inverted inequality or transcription slip separates the two chains. (b) Under Kourbatov's sharper constant the same computation yields 1922, so his published $g < 1920$ is conservative relative to his own criterion, and the exact coincidence with 1918 is partly arithmetic luck at even-gap granularity. (c) Lemma 4.2 yields the *local* statement at the frontier; the endnote's phrasing is *global*. Since $S^\sharp(1918) = 1.82 \cdot 10^{19}$, our bound leaves a gap of 1918 free to violate F anywhere below $1.82 \cdot 10^{19}$. Closing that needs the first-occurrence table again.

4.4 The table-free window, and its permanent closure

Can a range be certified with *no* prime enumeration at all? Only just, and only once.

Proposition 4.8 (Table-free window; [P·s])**.** F *holds at every n with* $396\,738 \leq p_n \leq 777\,600$*, with no computational input beyond two cited explicit estimates.*

Proof. Idea: Dusart's short-interval theorem caps g_n by $p/(25L^2)$; Lemma 4.2 floors T_n by $L^2 - 1.1L$; the two cross exactly once.

For $p_n \geq 396\,738$, [Dus10, Prop. 6.8] supplies a prime in $(x, x(1 + 1/(25 \ln^2 x))]$, hence $g_n \leq p_n/(25L_n^2)$. By Lemma 4.2, $T_n > L^2 - 1.1L$. So it suffices that

$$\frac{p}{25L^2} \leq L^2 - 1.1L \qquad \iff \qquad p \leq 25L^3(L - 1.1), \qquad L = \log p.$$

Let $h(p) := p - 25(\log p)^3(\log p - 1.1)$. Then $h(396\,738) = -2.35 \cdot 10^5 < 0$, and $h'(p) = 1 - (25/p)(4L^3 - 3.3L^2) > 0$ throughout $[4 \cdot 10^5, \infty)$, so h has a unique sign change there, at $p^* = 777\,600.744\ldots$ (§7, item V6). Hence the displayed inequality holds on $[396\,738, p^*]$, so $g_n < T_n$, so F at n by Lemma 2.1. ☐

Proposition 4.9 (The window closes permanently; [P]). *For $p > p^* = 777\,600.744\ldots$, [Dus10, Prop. 6.8] does not imply F, and the shortfall grows without bound: $\dfrac{p/(25L^2)}{L^2 - 1.1L} = \dfrac{p}{25L^3(L - 1.1)} \to \infty$.*

Proof. Idea: p outgrows every fixed power of $\log p$.

Immediate from the display in the proof of Proposition 4.8. ☐

Remark 4.10 (The precise measure of what analysis alone can do). A table-free proof of F on $[X_0, X]$ requires an unconditional explicit gap bound of quality $g_n = O(\log^2 p_n)$ on that range. Every known unconditional gap bound — Dusart's $p/(25 \log^2 p)$ here, $p^{0.525}$ of [BHP01], and everything between — is a *power of p*. So reading (B) is dead above $7.776 \cdot 10^5$ and will stay dead until unconditional prime-gap technology reaches $\log^2$ scale, which is exactly the strength that proving F itself requires. **Corollary, stated so that it is not re-derived: no amount of analysis will ever eliminate the computational input from a verified-range claim.** The verified range is irreducibly a computation plus a theorem about the computation. Theorem 4.4 is the theorem; §10 names the computation. The window $[396\,738, 777\,600]$ lies entirely above X_0 and is therefore already covered by (H1)–(H3) in any real verification; its value is that it is the exact measure of how much analysis alone can do, and the measure is: one factor of about 2 in p, once, at $p \approx 10^6$.

5 The Riemann-Hypothesis route is closed

The slogan "RH does not help with Firoozbakht" is common and usually unargued. This section replaces it with theorems. Five inequivalent readings of "an RH-conditional bound settles F" are separated; four are refuted, and the fifth is left undecided with a lower bound on the strength of what a proof would have to yield.

Throughout this section we use the CMS envelope

$$B_n := \tfrac{22}{25}\sqrt{p_n}\, L_n,$$

for which [CMS19, §1.2, Thm. 5] gives: RH implies $g_n \leq B_n$ for every $n \geq 3$ (equivalently $p_n > 3$; see [Vis18, Thm. 1, eq. (1.4)] for the same statement with hypothesis $n \geq 3$, $p_n \geq 5$).

We say the envelope *certifies* F at n when $B_n \leq T_n$; by Remark 2.2 this upgrades to the strict inequality $g_n < T_n$ that Lemma 2.1 demands.

We also use the following in-run fact, verified for all $n \leq 216\,815$ and asymptotically a consequence of [Dus10, Thm. 6.9]: $\{n : T_n \geq L_n^2\} = \{1, \ldots, 7, 10\}$.

Lemma 5.1 ([P]). *For every real $x > 0$, $\sqrt{x} > \frac{25}{22} \log x$; equivalently $\frac{22}{25}\sqrt{x}\, \log x > (\log x)^2$ for all $x > 1$.*

Proof. Idea: One stationary point, and it sits above zero.

Put $k := 25/22$ and $h(x) := \sqrt{x} - k \log x$ on $(0, \infty)$. Then $h'(x) = (\sqrt{x} - 2k)/(2x)$, so h has a unique stationary point at $x^* = (2k)^2 = 625/121 = 5.16528925\ldots$, a minimum, with $h \to +\infty$ at both ends. At that point

$$h(x^*) = \tfrac{25}{11} - \tfrac{25}{22} \log \tfrac{625}{121} = 0.40686238\ldots > 0.$$

Hence $h > 0$ throughout. Multiplying $\sqrt{x} > k \log x$ by $\log x > 0$ (valid for $x > 1$) and dividing by k gives the second form. □

Theorem 5.2 (The CMS envelope certifies at one index; [P])**.** *Let $S := \{n \geq 3 : B_n \leq T_n\}$. Then $S = \{3\}$. That is: the sharpest published RH-conditional prime-gap bound certifies the Firoozbakht inequality at the single index $n = 3$ ($p_3 = 5$) and at no other index in the range where the bound is available.*

Proof. Idea: Off a set of six indices the bar T_n is below L_n^2, and the envelope is above L_n^2 always; the six are checked by hand.

Let $n \geq 3$. If $n \notin \{3, 4, 5, 6, 7, 10\}$ then $T_n < L_n^2$, while Lemma 5.1 at $x = p_n \geq 5$ gives $B_n > L_n^2 > T_n$, so $n \notin S$. The six remaining indices are checked directly in 60-digit arithmetic:

n	p_n	T_n	B_n	$B_n < T_n$?
3	5	3.549879733	3.166955068	yes
4	7	4.386035932	4.530587009	no
5	11	6.769336928	6.998568638	no
6	13	6.934281081	8.138289656	no
7	17	8.481637825	10.27984133	no
10	29	11.61044961	15.95742984	no

All six comparisons are strict, so $\leq$ and $<$ agree on them. Hence $S = \{3\}$. □

Remark 5.3 (Read the quantifier). The restriction $n \geq 3$ in Theorem 5.2 is part of the statement and must not be dropped when the theorem is quoted. The envelope in fact clears the bar at $n = 1$ and $n = 2$ as well; what excludes them is the *source's* hypothesis $p_n > 3$, not the arithmetic. The correct informal form is therefore “at exactly one index in the range where the CMS bound is available”, never “at no other index whatsoever”.

Remark 5.4. The single certified index is worthless: $n = 3$ means F at $p = 5$, i.e. $7^3 = 343 < 5^4 = 625$, decidable by hand, and $n = 3$ falls below the threshold $k > 9$ that both Kourbatov criteria carry by hypothesis. Moreover RH is not even load-bearing where the envelope is smallest: by [Vis18, §7] the CMS inequality holds *unconditionally* for all $p < 1.836 \cdot 10^{19}$, essentially the whole verified range of Theorem 2.3 — and what is already known still certifies F at no index above $n = 3$.

Theorem 5.5 (No power-type envelope suffices; [P])**.** *Let $C > 0$, $\theta > 0$, $A \in \mathbb{R}$ and $E(x) := Cx^\theta (\log x)^A$. Then $\#\{n : E(p_n) \leq T_n\} < \infty$; indeed $E(p_n)/T_n \to \infty$.*

Proof. Idea: In the variable $u = \log x$ the envelope is exponential and the bar is quadratic.

Since $T_n < L_n^2$ for all $n \notin \{1, \ldots, 7, 10\}$, it suffices that $E(x)/(\log x)^2 \to \infty$. Write $u := \log x$; then $E(x)/(\log x)^2 = Ce^{\theta u} u^{A-2} \to \infty$ as $u \to \infty$, because $e^{\theta u}$ with $\theta > 0$ dominates any fixed power of u. Hence $E(p_n)/T_n > E(p_n)/L_n^2 \to \infty$, and a divergent ratio is ≤ 1 only finitely often. □

Every prime-gap upper bound currently available lives in this class: $p_n^{0.525}$ unconditionally [BHP01]; $O(\sqrt{p_n}\log p_n)$ under RH (Cramér 1919, as reported in [Vis18, §2, Thm. 4]); $4\sqrt{p_n}\log p_n$ under RH (Goldston 1982, [Vis18, §2, Thm. 5]); the explicit CMS bound and its asymptotic refinement $\limsup \le 1/C^+(B) < 21/25$ [CMS19, §1.2, Cor. 4]. **The column that matters is θ:** every entry has $\theta > 0$, while the bar (3) has $\theta = 0$. Improving the exponent from 0.525 to 0.5 — exactly what RH buys — moves nothing across the line, because the line is at $\theta = 0$.

Theorem 5.6 (The critical constant is $2/e$; [P]). *For $C > 0$ set $E_C(x) := C\sqrt{x}\log x$. Then $\{x > 1 : E_C(x) \le (\log x)^2\} \ne \emptyset$ if and only if $C \le 2/e = 0.735758882\ldots$, and when non-empty this set is the* bounded *interval with endpoints* $\exp(-2W_0(-C/2))$ *and* $\exp(-2W_{-1}(-C/2))$*, where W_0, W_{-1} are the two real branches of the Lambert W function. Consequently, for every $C > 0$, E_C certifies* F *at only finitely many indices.*

Proof. Idea: The condition is $C \le \log x/\sqrt{x}$, and that function is unimodal with maximum $2/e$ at $x = e^2$; solving the equality is a Lambert-W substitution.

$E_C(x) \le (\log x)^2 \iff C \le \varphi(x) := \log(x)/\sqrt{x}$ for $x > 1$. Now $\varphi'(x) = (2-\log x)/(2x^{3/2})$, so φ increases on $(0, e^2)$, decreases on (e^2, ∞), and attains its maximum $\varphi(e^2) = 2/e$ at $x = e^2 = 7.389056\ldots$. Hence the superlevel set $\{x > 1 : \varphi(x) \ge C\}$ is non-empty iff $C \le 2/e$ and, by unimodality with $\varphi \to 0$ at both ends, is a compact interval. Solving $C\sqrt{x} = \log x$ with $t = \log x$ gives $te^{-t/2} = C$, i.e. $(-t/2)e^{-t/2} = -C/2$, i.e. $-t/2 = W(-C/2)$, whence the stated endpoints; two real branches exist iff $-C/2 \ge -1/e$, i.e. $C \le 2/e$. Finally the set is bounded, so it contains finitely many primes, and away from it $E_C(x) > (\log x)^2 > T_n$ off the exception set $\{1, \ldots, 7, 10\}$. □

Both published constants, $22/25 = 0.88$ and $21/25 = 0.84$, are *above* the critical $2/e$, so both give the empty set. Constants far below it buy a bounded initial segment and then stop: $C = 0.1$ covers $p \in (1.111, 8099)$, and $C = 0.01$ covers $p \in (1.010, 2\,122\,265)$. To certify F merely up to the *already verified* frontier one would need $C \le (L^2-L-1)/(\sqrt{p}L) = 1.009\cdot 10^{-8}$ at $p = 2^{64}$ — not a constant improvement but the abandonment of the $\sqrt{p}$ scale.

Corollary 5.7 (Even $\limsup = 0$ does not help; [P]). *Suppose $g_n = o(\sqrt{p_n}\log p_n)$. This certifies* F *at no index.*

Proof. Idea: $o(\sqrt{p}\log p)$ does not even imply $g_n < \log^2 p_n$, so it cannot imply $g_n < T_n$.

The hypothesis is compatible with gaps far above the threshold: a sequence with $g_n = \sqrt{p_n}L_n/\log\log p_n$ satisfies $g_n/(\sqrt{p_n}L_n) \to 0$ while $g_n/L_n^2 = \sqrt{p_n}/(L_n \log\log p_n) \to \infty$. Killing the constant does not change the scale, and by Theorem 5.5 the scale is the whole obstruction. □

Caveat 5.8 (Provenance of the hypothesis of Corollary 5.7). That $\limsup = 0$ holds under RH together with Montgomery's pair correlation conjecture is *reported*, not proved, by [CMS19, §1.2, after eq. (1.14)], who attribute it to three works not opened in this study. It is therefore second-hand and is attributed as such here. Corollary 5.7 does not depend on the statement being true — it says that *even if* it holds, nothing is certified — so the weak tier does not propagate into the verdict.

5.1 The material implication RH $\Rightarrow$ F

Theorems 5.2–5.6 refute the *route*. They say nothing about the *proposition*. Conflating the two is the failure mode this section most needs to avoid, so we state plainly: to refute RH $\Rightarrow$ F one must establish RH $\wedge \neg\mathsf{F}$, i.e. prove the Riemann Hypothesis *and* exhibit a counterexample to F; to prove it one must derive F from RH. Neither is available and no computation bears on either. **The implication is undecided here and we do not pretend otherwise.** What can be proved is a lower bound on the strength of what such a proof would yield.

Theorem 5.9 ([P]). *Suppose* RH $\Rightarrow$ F *were proved. Then, as an immediate corollary, one would have proved*

$$\mathrm{RH} \implies \big(\forall k > 9 : g_k < L_k^2 - L_k - 1\big),$$

an RH-conditional Cramér-type prime-gap bound at $\log^2$ *scale, uniform in* k*, with leading constant* 1.

Proof. Idea: Compose with the unconditional necessary condition (2).

Compose the hypothesised implication with [Kou15a, §2, Thm. 1], which is the unconditional implication (2). □

Corollary 5.10 (The size of the required advance; [P]). *The ratio of the best published RH-conditional bound to the bound Theorem 5.9 would deliver is*

$$\frac{\frac{22}{25}\sqrt{p_n}\,L_n}{L_n^2 - L_n - 1} \longrightarrow \infty, \qquad \textit{the ratio growing like } \frac{22}{25}\cdot\frac{\sqrt{p_n}}{L_n},$$

with value $8.72 \cdot 10^7$ *at* $p = 2^{64}$ *and unbounded thereafter.*

Caveat 5.11 (No difficulty ordering is claimed). Corollary 5.10 is a *distance between two statements*, not a measure of proof effort. Nothing here says such a proof must first pass through a sharpened $\sqrt{p}$-scale bound, and nothing here orders RH $\Rightarrow$ F against any other open problem by difficulty. In particular this paper does *not* assert that F is harder than RH.

Corollary 5.12 (The circularity charge, made quantitative; [P·s]). *Let* H *be any hypothesis of gap-bound shape with* $H \Rightarrow \mathsf{F}$*. Composing with* (2) *gives* $H \Rightarrow (\forall k > 9 : g_k < L_k^2 - L_k - 1)$*. So every sufficient hypothesis is at least as strong as the necessary condition; and by* (3) *a hypothesis only an additive* 0.17 *stronger than that bound already suffices. Hence a candidate gap-bound hypothesis can sit only in the band between* $L^2 - L - 1$ *and* $L^2 - L - 1.17$*: outside it, a candidate is either too weak to give* F *or is itself a sufficient condition, i.e.* F *with a constant changed.*

Remark 5.13. Corollary 5.12 does *not* say that every sufficient H is within 0.17 of F — plainly false, since $g_k < L^2 - L - 5$ is sufficient and far stronger. The sandwich pins F, not the class of hypotheses that imply it. Corollary 5.12 uses (3), hence Axler (Caveat 3.5); the inequality *direction* is robust to the constant — any b with $1 < b < \infty$ gives the same conclusion with $b - 1$ in place of 0.17 — but the numeral 0.17 inherits that dependency. The corollary is also stated for hypotheses of gap-bound shape only.

5.2 Cramér's hypothesis does not entail F over integer sequences

The last escape is: assume Cramér's conjecture instead of RH. Write

$$\text{(Cr)} \qquad \limsup_{n\to\infty} g_n/L_n^2 \;\le\; 1. \tag{5}$$

Lemma 5.14 ([P]). *For every* $\delta > 0$ *there are infinitely many* n *with* $g_n < (1+\delta)L_n$.

Proof. Idea: If all gaps were eventually $\ge (1+\delta)\log p_n$*, the primes would be sparser than the prime number theorem allows.*

Suppose not: $g_n \ge (1+\delta)L_n$ for all $n \ge N_0$. Summing, $p_N \ge (1+\delta)\sum_{n=N_0}^{N-1} L_n$. By the prime number theorem $p_n \sim n\log n$, so $L_n \sim \log n$ and $\sum_{n\le N} L_n \sim N\log N$. Hence $p_N \ge (1+\delta+o(1))N\log N$, contradicting $p_N \sim N\log N$ for large N. □

Theorem 5.15 (Counter-model; [P])**.** *Assume* (Cr)*. Then there is a strictly increasing sequence* $(q_n)_{n\geq 1}$ *of positive integers with*

1. $q_n = p_n + O\big((\log n)^2 \log\log n\big)$;
2. $\limsup_{n\to\infty}(q_{n+1} - q_n)/(\log q_n)^2 = 1$, *so* q *satisfies* (Cr);
3. $q_{n+1}^{1/(n+1)} \geq q_n^{1/n}$ *for infinitely many* n*, so* q *violates the Firoozbakht inequality infinitely often.*

Hence (Cr) *does not entail* F *in the theory of strictly increasing sequences of positive integers.*

Proof. Idea: Take the primes and, at a sparse sequence of indices where the gap happens to be small, inflate that one gap up to exactly $\lceil \log^2 \rceil$*; the inflations are so rare and so small that they do not disturb the* lim sup *or the density, but each one breaks the Firoozbakht inequality.*

By Lemma 5.14 with $\delta = 1$, for each $k \geq 1$ let $n_k := \min\{n \geq 2^{2^k} : g_n \leq 2L_n\}$, which exists, and $n_k \to \infty$. Define recursively

$$J_k := \lceil (\log q_{n_k})^2 \rceil - g_{n_k}, \qquad q_n := p_n + \sum_{k\,:\,n_k<n} J_k;$$

the recursion is well founded because q_{n_k} depends only on $J_1, \ldots, J_{k-1}$.

Positivity and monotonicity. $g_{n_k} \leq 2L_{n_k}$ while $\lceil (\log q_{n_k})^2 \rceil \geq L_{n_k}^2$, and $L^2 > 2L$ for $L > 2$, i.e. $p > e^2$. Start the construction at the least k_0 with $p_{n_{k_0}} \geq 11$ and discard $k < k_0$; this is legitimate because the conclusion concerns infinitely many indices, and it makes every $J_k \geq 0$. Then q is p plus a nondecreasing step function, hence strictly increasing.

Item 1. $q_n - p_n = \sum_{k:n_k<n} J_k$ with $J_k \leq \lceil (\log q_{n_k})^2 \rceil$. Since $n_k \geq 2^{2^k}$, the number of terms with $n_k < n$ is at most $\log_2 \log_2 n + O(1)$, and each $J_k \leq (\log q_n)^2 + 1 = O((\log n)^2)$. Hence $q_n - p_n = O((\log n)^2 \log\log n)$.

Item 2. At $n = n_k$, $q_{n_k+1} - q_{n_k} = g_{n_k} + J_k = \lceil (\log q_{n_k})^2 \rceil$ by construction, so the ratio is $\lceil (\log q_{n_k})^2 \rceil / (\log q_{n_k})^2 \to 1$. At $n \notin \{n_k\}$, $q_{n+1} - q_n = g_n$ and $\log q_n \geq \log p_n$, so the ratio is at most g_n/L_n^2, whose lim sup is ≤ 1 by (Cr). Hence the lim sup is exactly 1.

Item 3. Write $T_n^{(q)} := q_n(q_n^{1/n} - 1)$; the derivation of Lemma 2.1 is purely formal and applies to any strictly increasing positive sequence, so the Firoozbakht inequality fails at n iff $q_{n+1} - q_n \geq T_n^{(q)}$. The index n is unchanged by the construction, and $q_n = p_n(1 + \varepsilon_n)$ with $\varepsilon_n = O((\log n)^2 \log\log n/p_n)$, which tends to 0 faster than any negative power of L_n. Since $\partial T_n/\partial p_n = O(L_n^2/p_n)$, the perturbation moves the bar by $O(\varepsilon_n L_n^2) = o(1)$; with $T_n = L_n^2 - L_n - 1 + o(1)$ [Kou15a, §5, Thm. 5] this gives $T_n^{(q)} < (\log q_n)^2$ for large n. At $n = n_k$ the gap is $\lceil (\log q_{n_k})^2 \rceil \geq (\log q_{n_k})^2 > T_{n_k}^{(q)}$. So the inequality fails at every sufficiently large n_k. □

Remark 5.16 (What Theorem 5.15 does and does not close). It closes the reading "Cramér's lim sup hypothesis entails F" *as a counter-model result.* It does *not* prove (Cr) $\not\Rightarrow$ F as a material implication about the primes: p is one fixed sequence, and if both (Cr) and F happen to be true the implication is vacuously true. Theorem 5.15 quantifies over sequences, on purpose. The stronger informal reading — "no derivation using only growth and distribution can succeed" — is a *gloss*, not a corollary: the drift in item 1 is small enough that no unconditional $\pi(x)$ estimate in our toolbox separates q from the primes (the induced displacement of the counting function is $O(\log x \log\log x)$, against a Dusart bracket of width $0.2762\,x/\log^2 x$), so a derivation appealing only to (Cr), the prime number theorem and such estimates would have to prove F for q as well. That argument is only as good as its unformalised premise about which appeals a derivation makes, and the class it names is contingent on what this study fetched. **Quote the theorem, not the gloss.**

Structurally, Theorem 5.15 explains why the additive 0.17 of Corollary 5.12 is not a technicality: (Cr) controls a $\limsup$, whereas F needs a uniform bound at *every* index with the second-order term pinned. The distance between "asymptotically at most 1" and "below $L^2 - L - 1$ always" is exactly the room the construction lives in.

6 The smooth model

It is worth recording why the difficulty is entirely in the fluctuation.

Proposition 6.1 ([P]). *The function $x \mapsto (x \log x)^{1/x}$ is strictly decreasing on $x \geq 5$.*

Proof. Idea: Take logarithms; the derivative of $\log(x \log x)/x$ has the sign of $1 + 1/\log x - \log(x \log x)$, which is negative once x is large enough, and $x = 5$ is large enough.

Write $\psi(x) := \log(x \log x)/x$. Then $x^2\psi'(x) = \bigl(1 + 1/\log x\bigr) - \log(x \log x)$. The bracket $1+1/\log x$ decreases and $\log(x \log x)$ increases, so ψ' has at most one sign change; evaluating, the bracket at $x = 5$ is 1.6213 while $\log(5 \log 5) = 2.0853$, so $\psi'(5) < 0$, and $\psi' < 0$ for all $x \geq 5$. Hence ψ, and therefore $(x \log x)^{1/x} = e^{\psi(x)}$, is strictly decreasing on $[5, \infty)$. $\square$

Since $p_n \sim n \log n$, Proposition 6.1 says that the *smooth model* of Firoozbakht's conjecture is true and elementary. The entire difficulty of F localises in the fluctuation of p_n around $n \log n$. The safe stated range is $x \geq 5$, not $x \geq 4$: the relevant bracket changes sign just above 4 ($+0.0084$ at $x = 4$, -0.0193 at $x = 4.05$). Proposition 6.1 was *not* formalised in Lean in this work (§8).

7 Computational evidence

7.1 Design

Two independent legs, with independent code paths, swept the primes exhaustively to 10^{11} — 4 118 054 812 consecutive prime pairs, no pruning, no sampling, no assumed lemma. The protocol is two-tier and the tiering is the point:

- **Screen.** `float64` over all pairs, on $\rho_n = g_n/T_n$. This statistic has $O(1)$ operands and so does not carry the catastrophic cancellation that afflicts the ratio statistic of §7.4.
- **Escalation.** Every pair with $\rho_n \geq 0.90$ is re-decided in arbitrary precision with an explicit error budget, and the certified verdict function *raises* rather than returns when the margin lands inside that budget.
- **Calibration.** The high-precision path is checked against the exact integer criterion $p_{n+1}^n < p_n^{n+1}$ for every $n \leq 5000$ in one leg and $n \leq 4000$ in the other. Zero disagreements.

The escalation discipline deserves emphasis because it inverts the usual default: *the silent failure here is in the verification direction.* A probe that only breaks on a detected violation reports "no counterexample" out of numerical noise. This one cannot.

7.2 Results

Quantity	Value
consecutive prime pairs checked ($p_n < 10^{11}$)	4 118 054 812
violations of F	**0**
$\max_{n\geq 10} \rho_n$	**0.8317570** at $p_n = 25\,056\,082\,087$, $g = 456$
$\max_{n\geq 10} g_n/L_n^2$	0.7953487, same prime
largest gap in range	464 at $p = 42\,652\,618\,343$
maximal-gap records in range	40
sieve validation	$\pi(10^9)$, $\pi(10^{11})$ match standard values

Only two pairs in the whole range reach $\rho \geq 0.9$, and both are at $n < 5$ ($\rho_2 = 0.9107$, $\rho_4 = 0.9120$), which is why the record is quoted for $n \geq 10$.

7.3 Verification of the lemmas of §3–§4

The following items are cited by number from the proofs above. All were recomputed independently of the code paths that produced the sweep.

V1 $\max\{g_k : p_k < 60\,184\} = 72$, attained at $p = 31\,397$, by exact integer arithmetic over the 6 076 gaps below 60 184; and $B(60\,184) = 109.008\ldots > 72$. Also: $p_{n+1}^n < p_n^{n+1}$ for every one of those 6 076 indices in exact integer arithmetic — this is hypothesis (H1) of Theorem 4.4, discharged.

V2 $S(29) = 6.8013853766\ldots > 6 = \max\{g_j : j \leq 9\}$ — hypothesis (ii) of Theorem 3.6.

V3 Lemma 4.2 holds at every prime in $[60\,184, 10^{11}]$; the tightest slack is $T_n - B(p_n) = +0.0798914728\ldots$, attained at the very bottom of the range ($p = 155\,893$).

V4 Lemma 4.1 holds at every prime in $[60\,184, 2 \cdot 10^7]$; zero failures.

V5 $B(2^{64}) = 1919.1379834975328$ (Proposition 4.6), and the largest even g with $S^\sharp(g) \leq 2^{64}$ is 1918.

V6 The sign change of $h(p) = p - 25(\log p)^3(\log p - 1.1)$ is at $p^* = 777\,600.744\ldots$ (Proposition 4.8).

V7 Table lookup for the whole in-run range: of the 209 gap sizes occurring below 10^{11}, 55 are excluded outright by $S^\sharp(g) < 60\,184$ and 154 need the table test; **none is unsettled**. The minimum safety factor $q(g)/S^\sharp(g)$ is 5.337 at $g = 112$, the maximum 203.9 at $g = 334$.

V8 At the published record Cramér–Shanks–Granville point $p = 1\,693\,182\,318\,746\,371$, $g = 1132$ [Sloa]: $B(p) = 1191.009$, so the record gap is safe with a margin of 4.955%.

V9 The certificate that scales: splitting $[60\,184, 10^{11}]$ into 22 geometric windows $[a, b)$ and checking $\max\{g_n : p_n < b\} < B(a)$ certifies every window, consuming *only* the 40 maximal-gap records — no sieve, no $\pi(x)$ computation, no per-prime check. This is the shape in which a frontier like 2^{64} is actually certified.

Remark 7.1 (Stability of the table test across four decades). The minimum safety factor $q(g)/S^\sharp(g)$ is 5.3371 at $g = 112$ at every one of $X = 10^7, 10^8, 10^9, 10^{10}, 10^{11}$. Four decades of new data and 134 new gap sizes produced nothing tighter than a case already visible in the first decade ($g = 112$, first occurring at $p = 370\,261$). Route (b) neither tightens nor loosens as the range extends.

7.4 Three refutations of claims about the evidence

Computation refutes nothing about F. It did refute three claims about the evidence, and recording them is the point of this subsection.

(1) The "0.9999984 near-miss" is an artefact. The statistic $\max_n n \log p_{n+1}/((n+1)\log p_n)$ reads 0.9999984 below $3 \cdot 10^6$ and has been read as evidence that F nearly fails. It is not. In a synthetic universe where every gap equals 2 (so that F holds by a mile) the same statistic reads 0.99999991 at $n = 10^7$, while ρ_n there is 0.059 — an order of magnitude below the bar. The statistic measures $1/n$. Only ρ is diagnostic:

X	10^7	10^8	10^9	10^{10}	10^{11}
max ratio statistic	0.99999937	0.99999994	0.999999993	0.9999999992	0.99999999992
$\max_{n\geq 10} \rho_n$	0.7605	0.7896	0.7896	0.7896	0.8318

(2) "The tightest cases sit at record gaps" does not survive the range. The observation that the tightest ρ cases occur at record gaps was a print truncation of the first six entries. At 10^{11}: 22 of the top 40, and the fourth-tightest case among four billion pairs is *not* at a record index ($g = 444$ at $p = 36\,172\,730\,063$, $\rho = 0.78502$). Read precisely: if the pruning rule holds, a record-index search still misses no *counterexample*; what it misses is *near-misses*. So the observation cannot be used as evidence for the pruning rule that produced it.

(3) More sieve is not more evidence. Two decades of extra sieving ($10^8 \to 10^{10}$) produced no new near-miss: the record ρ sat at 0.7896 ($g = 210$, $p = 20\,831\,323$) throughout. It moved exactly once, at 10^{11}, and it moved *at the next maximal gap*. Sieving buys the next maximal gap and nothing in between. The record that would matter — $\rho \approx 0.948$ at $p \approx 1.693 \cdot 10^{15}$ [Sloa] — is 4.2 decades above anything reached here. The compute that would matter goes into gap *tables*, not prime *counts*.

7.5 A counter-model for first-failure maximality

Proposition 7.2 ([C])**.** *There is an explicit strictly increasing sequence of positive integers whose first Firoozbakht failure occurs at a non-record gap.*

Proof. Idea: Take the real primes up to 1327*, then a large jump, then a long dense run of gaps of* 2*, then a jump smaller than the first one; the dense run drives the index up so far that the later, smaller gap is the one that breaks the inequality.*

Take the primes up to 1327, then a jump of 40, then 100 jumps of 2, then a jump of 38. The first Firoozbakht failure of this sequence is at index 323 on the gap of 38, which is not a record (the earlier jump of 40 is larger). Verified in exact integer arithmetic, so rounding cannot be blamed. □

The consequence is sharp: **any proof of first-failure maximality that does not consume an arithmetic density input is wrong**, because the statement is false for sequences satisfying everything except primality. The good news is that a weak input suffices for this mechanism: the dense run the counter-model needs exceeds the unconditional Montgomery–Vaughan cap $2y/\log y$ by a factor that *grows* with scale — 1.41 at $p \approx 10^3$, 4.18 at $4 \cdot 10^8$, 9.13 at 10^{18} — so the counter-model is excluded unconditionally, and increasingly comfortably. (The Montgomery–Vaughan bound is not covered by a source-ledger row in this work; cf. Caveat 3.13.)

A second, independent route is also closed: bounding T_m above and T_n below with explicit $\pi(x)$ estimates *at the two endpoints separately* demands record gaps of size at least $5.3 \cdot 10^4$ at $p = 10^6$ (Dusart) or $3.7 \cdot 10^3$ (Axler), against actual records of size $\approx \log^2 p \approx 176$; and past $\approx 10^{10}$ (Dusart) the criterion is unsatisfiable at *any* gap size. The bound spread dwarfs the quantity being resolved, and no sharpening of constants rescues the route.

8 Formalisation in Lean 4

8.1 What is machine-checked

The reduction chain

$$\texttt{Conjecture} \leftrightarrow \texttt{ConjectureReal} \leftrightarrow \big(\forall n \geq 1,\ g_n < T_n\big)$$

— i.e. Lemma 2.1 — is formalised in Lean 4 against Mathlib and checked by the kernel. Toolchain: `leanprover/lean4:v4.29.0`; Mathlib tag `v4.29.0`, revision `8a178386ffc0f5fef0b77738bb5449d50efeea95`.

Gate	Result
`lake build` exit code	**0** (1984 jobs)
build warnings	**1** — the declared open target, and nothing else
declarations scanned by the axiom audit	**60** (exhaustive, list-free)
declarations depending on `sorryAx`	**1** — `Firoozbakht.firoozbakht`
live `sorry` tokens in sources	**1**
stray `axiom`, `native_decide`, `@[implemented_by]` or `unsafe`	**none** (grep-clean)

The single `sorryAx` dependent is Firoozbakht's conjecture itself, which was never attempted. *An attack on an open Π_1 statement whose `sorry` count reaches zero would be reporting a fabrication, not a result.* Four of the skeleton's original five `sorry`s — the equivalence-chain lemmas — were discharged with real proof terms, and the signatures were mechanically diffed against the skeleton anchor with zero changes, which is what forecloses the failure mode "closed a `sorry` by weakening the statement". The equivalences were also shown non-vacuous: $g_1 = 1$ and $T_1 = 2^2 - 2$ are pinned to numerals.

The audit itself was made exhaustive rather than trusted: a hand-maintained list of declarations to `#print axioms` is a hazard dressed as a check, since a `sorry` in a forgotten declaration is invisible. The list-free audit walks the namespace. The checker was then tested against a corpus of 27 adversarial statements, all false or ill-formed; all 27 behaved as specified. One corpus entry — a true-but-differently-meant statement produced by an $\mathbb{N} \to \mathbb{R}$ coercion — is caught by *no* gate in this study and is named as such rather than omitted.

8.2 What is not formalised, and why

- **The smooth model** (Proposition 6.1). Fully within Mathlib's reach; simply not done.
- **The** $\limsup$ **corollary.** Needs effective $\pi(x)$ bounds that are not assumed present in Mathlib.
- **Any verified range past** $n \leq 4$**.** This is a hard limit and deserves naming. `Nat.nth` is *noncomputable* [Thea] — not merely slow — so `decide` cannot produce p_n at all; and Mathlib's prime-specific `nth` API is exactly five `@[simp]` base lemmas, `nth Prime` $0 = 2$ through `nth Prime` $4 = 11$ [Theb]. Extending requires building `Nat.count`↔`Nat.nth`

bridging machinery, plus prime literals with `Nat.Prime` certificates and a "no prime strictly between" lemma for each consecutive pair. Reporting a larger verified N without that work would be a fabrication.

- **Dusart's estimate** (Lemma 4.1) is not in Mathlib and would have to enter any formalisation of §4 as a hypothesis or an axiom — and be declared as such in any paper quoting the result.

Caveat 8.1 (Indexing). Mathlib's `nth` is 0-indexed (`nth Prime 0` $= 2$) while the conjecture as posed here is 1-indexed ($p_1 = 2$). Every index threshold in the literature — $k > 9$, $n \geq 10$, $n \geq 3645$, $n \geq 5$ — is stated in the 1-indexed convention. The off-by-one must be fixed once, in the statement file; it is the single most likely source of a silent error in a formalisation of this material.

9 The tension that will not resolve itself

The strongest reason to believe F is *false* is not a computation and not a theorem. Cramér's model, in Granville's corrected form, predicts $\limsup g_n/\log^2 p_n \geq 2e^{-\gamma} \approx 1.1229 > 1$ [Gra95, preprint p. 12], which is incompatible with F via (2). This is [H]: a heuristic, not a test. Granville's own sign is $\gtrapprox$ — "suggests", not "proves".

Symmetrically, the strongest reasons to believe F is *true* are the verified range (Theorem 2.3) and the fact that the record Cramér–Shanks–Granville ratio 0.9206, standing since 1999 [Sloa], sits comfortably below the Firoozbakht-implied ceiling of $1 - 1/\log p \approx 0.9715$ at that size — about 5% of headroom. Neither is a proof and neither is close to one.

Caveat 9.1 (Cramér did not conjecture $\limsup = 1$ about the primes). He *proved* it about his urn model, with probability 1 [Cra36, p. 27], and *suggested* the analogue for the primes [Cra36, p. 24, eq. (4)]. Granville frames it the same way. Any sentence of the form "Cramér conjectured that $\limsup g_n/\log^2 p_n = 1$" is a compression the source does not literally support.

10 What remains open, and what this paper does not establish

This section is long on purpose. A paper reporting an attack that failed to settle its target is exactly the paper that gets over-quoted downstream.

10.1 F itself, and why it is not close

The load-bearing obstruction is unchanged by anything above: **any proof of F yields $g_n = O(\log^2 p_n)$ unconditionally**, via (2). The best known unconditional gap bound is $g_n = O\big(p_n^{0.525}\big)$ [BHP01]; under RH it improves only to $\approx \sqrt{p_n}\log p_n$ [CMS19]. Both are powers of p_n; F needs polylogarithmic. Any proposal of a "direct proof" must say how it clears this, or it is proposing something already beyond the field.

Compounding it: **there is no induction mechanism.** g_n is not constrained by $g_1, \ldots, g_{n-1}$; knowing F up to n gives no leverage at $n + 1$. Any proposed inductive proof must first supply the missing mechanism.

On the refutation side, the best large-gap results [For+16] reach an iterated-log factor above $\log n$ but remain a full power of log below what a counterexample needs. Explicit constructions fail for a second, independent reason: they place the gap at an *unspecified* location, while F needs the gap and the count $\pi(p_n) = n$ at the *same* point. Correspondingly, although $\neg\mathsf{F}$ is Σ_1 and finitely certifiable in principle, the certificate must certify the *rank* n, not merely the two primes.

10.2 Named defects in the underlying corpus

The body of work this paper draws on was subjected to an adversarial review that found two unrepaired blocking defects. Neither touches F, and both are disclosed here rather than smoothed.

1. **A vocabulary collision on the "governing record index".** The phrase carries three inequivalent meanings across the corpus, and the same statistic was reported under all three: (i) $m(n)$ = the most recent maximal-gap index $\le n$; (ii) $m(n) := \min\{m : g_m \ge g_n\}$; (iii) $T_m \le T_n$ for *all* $m < n$ straddling a record gap. These are strictly ordered, (iii) $\Rightarrow$ (i) $\Rightarrow$ (ii), and the search pruning consumes only the weakest, (ii). With that named, two apparently contradictory findings cease to contradict: the margin of predicate (i) decays like $p^{-0.83}$ ($+1.05 \cdot 10^{-2}$ at $3 \cdot 10^6$ to $+2.93 \cdot 10^{-6}$ at 10^{11}, zero exceptions throughout), while the margin of predicate (ii) is flat at $+0.4845$ with its global minimum at $n = 1879$, never approached again across eleven decades, also with zero exceptions. **This paper is written entirely in the disambiguated vocabulary**, but the upstream artifacts still carry the collision, and an inference drawn there — that a `float64` check of the pruning predicate hits its noise floor at the published frontier — is derived from predicate (i) and does *not* apply to predicate (ii), which is $\sim 10^{12}$ ulps clear of zero at every decade measured.
2. **A mis-derived bound in the Axler-sharpened version of Theorem 3.10.** As printed upstream, the theorem is false by a factor ≈ 38 over part of its own validity range, because the stated upper bound on T_m does not follow from its stated justification; the in-run numerical check reproduced the error rather than catching it. The review confirmed that the *conclusion* is nonetheless true under the corrected tight form, and the repair is a one-line restatement. **It has not been applied.** This paper therefore states only the Dusart-only Theorem 3.10, whose constant 0.0623 is unaffected and was independently confirmed conservative; see Remark 3.11.

Consequently the pre-publication evidence gate on the underlying corpus stands at **BLOCKED**, with the adversarial-review leg named as the failing one. The formal (kernel) leg passes outright, and the adversarial-corpus leg passes (27/27). Nothing in §§2–9 rests on either defect.

10.3 Source provenance, and the citation audit that has not run

Every citation in this paper traces to a row of a source ledger built from scratch for this study: 20 rows, of which 11 were fetched and read at the recorded locator, 3 were fetched at a pagination that could not be independently confirmed, 4 are attested at a specific locator by two or more independent fetched primary sources, and 2 by one such source. No citation rests on recall. Seven source PDFs were fetched and read in full with their MD5 digests recorded.

Two entries in `references.bib` — [CMS19] and [Vis18] — are **pending ledger rows**: they were fetched, version-pinned and read at the locator during this study, and proposed as ledger additions, but were not folded into the ledger artifact. They are flagged here and in `authoring-log.md`. Locators for [CMS19] are to arXiv:1708.04122v2 and cannot presently be mapped to the journal's numbering.

The citation audit for this paper has not been run, and no citation clearance is claimed. Three unopened or under-located sources are load-bearing, in priority order:

1. [Axl14] — *not opened.* Quoted through Kourbatov's proofs; Kourbatov's Theorems 1, 3 and 5 depend on it, and so do Lemma 3.4, Theorem 3.6 and Corollary 5.12. Axler's

own corrigendum moved the validity range of Corollary 3.5 by nine orders of magnitude. **Priority 1.**

2. [Gra95] — fetched and read, but at *preprint* pagination (pp. 1–16) against the journal's pp. 12–28. Every locator must be re-expressed against the journal copy. This is the load-bearing citation of the entire refutation-side argument (§9). **Priority 2.**

3. [OHP14] — *not opened* (the publisher returned HTTP 403). It is the first-occurrence gap table on which Theorem 2.3 rests, and therefore the single largest external dependency of every statement about 2^{64} in this paper. If that table is incomplete, Proposition 4.6 and every use of Theorem 2.3 say nothing. **Priority 3.**

Further, [Rib04] (the origin locator, p. 185) was not opened and rests on two independent citers; [Sha64] and [FJ17] were not opened and are cited for attribution only, their content being available at first hand elsewhere in the bibliography; and [BHP01] and [For+16] were read at abstract level only — adequate for the qualitative use both are put to, inadequate if any internal constant were quoted, which it is not.

10.4 Scale, stated plainly

The in-run exhaustive sweep reached $p < 10^{11}$, which is 8.27 decades short of 2^{64}. Several of the lemma verifications were run at the smaller scale $3 \cdot 10^6$, which is 12.8 decades short of 2^{64}. **Both sweeps verify the lemmas, not the range.** They must never be cited as a verification of F. The distance between what this study owns and the published frontier is now exactly one external table and one inequality to check against it: for every even g occurring as a prime gap in $[10^{11}, 2^{64}]$, is $q(g) > S^\sharp(g)$? The checker is written and validated on data we own; supply the table and the frontier is re-derived in seconds.

Finally: "unconditional" means "no unproved hypothesis", not "no computation". Proposition 4.9 proves the computation cannot be removed. Any sentence of the form "F is unconditionally true below X" must, to be honest, carry the table with it.

10.5 Dead ends, recorded so they are not rediscovered

- **Littlewood oscillation is irrelevant** to the threshold, its contribution being $O\big(L^2 \log\log\log x/\sqrt{x}\big)$, which tends to zero.
- **Bertrand's postulate is useless** here: it implies F only at $n = 1$.
- **The two-sided π-bound route** to the maximal-gap reduction cannot work (§7.5).
- **More sieving is not more evidence** (§7.4(3)). The record that would matter is 4.2 decades above anything reached here.

11 Conclusion

Firoozbakht's conjecture was neither proved nor refuted here. What was produced instead is a machine-checked reduction of it to a prime-gap inequality; an unconditional, table-driven verification architecture that reproduces the published 2^{64} frontier from first principles and rests on a single explicit estimate for $\pi(x)$; an exhaustive independent sweep to 10^{11} finding no counterexample and no near-miss; four theorems closing the Riemann-Hypothesis route as a route; and the discharge, without proving it, of the obligation this attack had ranked first — offset by two named, unrepaired defects in the underlying corpus and by a citation audit that has not run.

The most tractable genuinely open analytic node isolated here is the residual window of Theorem 3.10: an unconditional short-interval prime count sharp to a factor $1 + 2/\log p$, where Brun–Titchmarsh gives only a factor 2 (Proposition 3.12). It is stated in closed form. It is also still an open problem in analytic number theory, not a lookup.

The standing instruction with which this work was carried out is the one it ends on. *The conjecture is open. Do not write "Firoozbakht is true". Do not write "Firoozbakht is false". The Cramér–Granville tension is evidence about which way to bet, and nothing more.*

Acknowledgements and provenance

This paper was produced by **Noogram** as the editorial artifact of a structured, multi-leg attack conducted on 25 July 2026. The legs contributing results reported here are: a decomposition leg; a source-ledger leg; a concept-card leg; three proof-attempt legs; three computational-notebook legs; Lean skeleton and probe legs; an adversarial red-team corpus leg; an adversarial review (skeptic) leg; an evidence gate; and a synthesis leg. The delivery posture of this artifact is **staged**: it reports a corpus whose evidence gate stands at BLOCKED (§10.2) and whose citation audit has not run (§10.3). It is not a seal.

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